Corporate Finance — Problem Set 1

Answer key (full derivations in `PS1-solutions.pdf`)

Exercise 1

(a) Value is the present value of the *cash* investors receive, dated when they receive it; net income is an accrual number. The differences:

$$FCF = NI + (1-\tau_c)Int + Dep - CAPEX - \Delta NWC.$$

Net income (i) deducts depreciation, which is not cash; (ii) ignores capex when it is paid, spreading it over time as depreciation; (iii) ignores cash tied up in working capital (sales on credit, inventory, payables); (iv) is after interest, so it mixes financing with operations. Financing is handled in the discount rate (WACC), so FCF is computed before payments to any investor.

(b) Yes. FCF must use taxes actually paid. Since $T^{cash} = T^{book} - \Delta DTL$,

$$FCF_t = EBIT_t(1-\tau_c) + Dep_t + \Delta DTL_t - CAPEX_t - \Delta NWC_t.$$

The deferred tax is a non-cash expense, like depreciation. Use only operating deferred taxes (a rise in deferred tax *assets* is subtracted). Do not also subtract the DTL as debt when moving from enterprise to equity value, and model the reversal or steady growth of the DTL in the terminal value. Equivalently, compute taxes on EBIT with the tax-depreciation schedule.

Exercise 2 (thousands of dollars)

Tax rate = 2,625/7,500 = 35%. The 1,000 consultancy fee is sunk: exclude it. Only 1,000 of overhead is incremental.

- $EBIT = 30{,}000 - 18{,}000 - 1{,}000 - 2{,}500 = 8{,}500$; $EBIT(1-0.35) = 5{,}525$.
- Years 1–9: $FCF = 5{,}525 + 2{,}500 = 8{,}025$.

(a) $FCF_0 = -25{,}000 - 10{,}000 = -35{,}000$; $FCF_1 = \cdots = FCF_9 = 8{,}025$; $FCF_{10} = 8{,}025 + 10{,}000 = 18{,}025$.

(b) With $A(0.14, 10) = 5.216116$ and $1.14^{-10} = 0.269744$:

$$\text{NPV} = -35{,}000 + 8{,}025(5.216116) + 10{,}000(0.269744) = 9{,}556.77.$$

The project is worth about \$9.56 million: accept (IRR $\approx 20.2\%$).

Exercise 3 (dollars)

- Old machine's book value: $110{,}000 - 10{,}000 = 100{,}000$. Selling at 50,000 gives a loss with tax saving $0.45 \times 50{,}000 = 22{,}500$, so the after-tax sale is 72,500.
- $\Delta FCF_0 = -150{,}000 + 72{,}500 = -77{,}500$.
- Years 1–10: $\Delta EBITDA = 20{,}000$, $\Delta Dep = 15{,}000 - 10{,}000 = 5{,}000$, so $\Delta FCF = 20{,}000(0.55) + 0.45(5{,}000) = 13{,}250$.

$$\Delta\text{NPV} = -77{,}500 + 13{,}250 \times A(0.10, 10) = -77{,}500 + 13{,}250(6.144567) = 3{,}915.51 > 0.$$

Yes, replace (incremental IRR $\approx 11.2\% > 10\%$). Without the tax saving on the sale the NPV would be −18,584.49: the tax effect of selling below book value decides the answer.